

# Applied Electronics 應用電子學

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Electronics and Photonics Post-undergraduate Second Degree Program  
National Yang Ming Chiao Tung University

# Syllabus

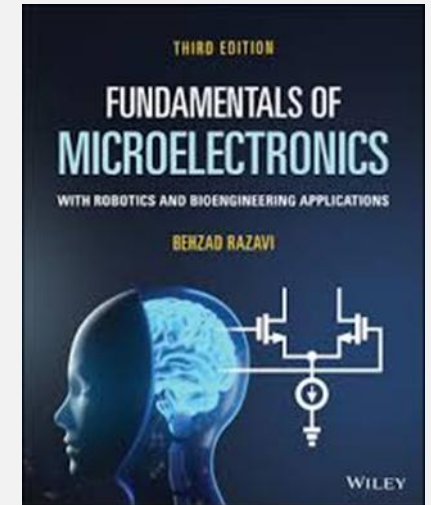
This course covers:

- (1) The operating principles and circuit models of basic semiconductor devices, including diodes and transistors.
- (2) The design, analysis, and applications of amplifier circuits.

Its goal is to equip students with the ability to design various circuits using basic semiconductor devices.

# Textbook

- Textbook:  
Behzad Razavi, Fundamentals of Microelectronics, 3rd Edition, Wiley, 2021. (滄海書局)
- References  
[1] Adel S. Sedra, Kenneth C. Smith, Tony Chan Carusone, and Vincent Gaudet, Microelectronic Circuits, 8th Edition, Oxford University Press, 2020.
- Prerequisites: No

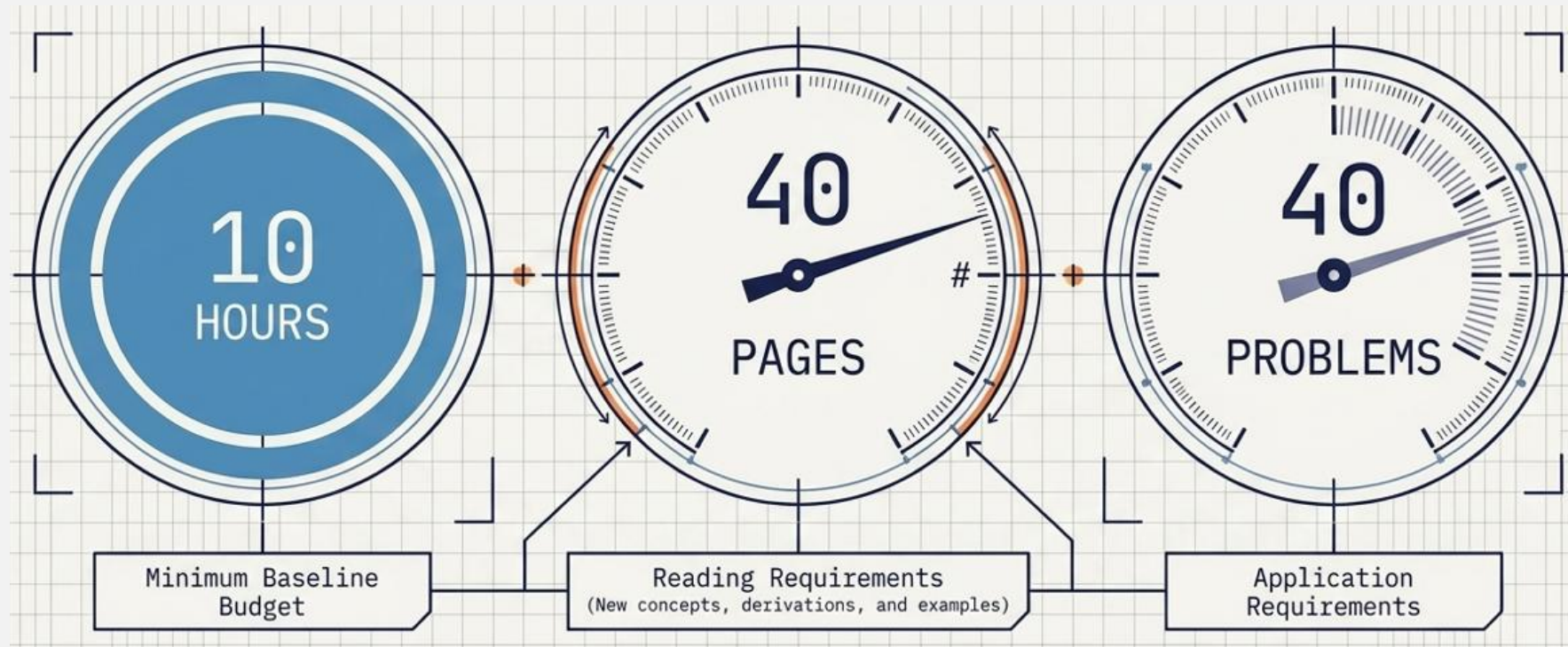


# Instructor

- Instructor: 林尚亭
  1. Tel: (07)3461666 # 78820
  2. E-mail: stl@nycu.edu.tw
- Grading:
  1. Quizzes, HW and Labs: 50%,
  2. Midterm: 20%,
  3. Final: 20%,
  4. Final Project: 10%

# Suggestion for Students

- For students, the path to success involves high-volume active learning (the "40/40 rule") and disciplined time management.



# Suggestion for Students

- **Mathematical Rigor:**  
Maintaining the precision required for engineering calculations.
- **Engineering Intuition:** (Hard work required)  
Developing a "feel" for how microelectronic devices and circuits operate. Without this intuitive "map," the analysis of complex circuits becomes increasingly difficult as more devices are added.
- **Analysis by Inspection**  
**Decomposition:** Breaking down complex circuits into simpler, recognizable topologies.  
**Algebraic Formulation:** Describing circuit behavior with a few lines of algebra rather than exhaustive, inefficient calculations.

# Schedule – Applied Electronics

- Introduction to Microelectronics, Basic Physics of Semiconductors
- Diode Models and Circuits
- EDA Tools & Simulators
- Lab: Half-wave Rectifier, Full-wave Rectifier
- Physics of Bipolar Transistors
- Bipolar Amplifiers
- Lab: CE Amplifier
- MOS Amplifiers
- Lab: CS Amplifier
- Midterm
- Operational Amplifier
- Lab: Inverting Amplifier, Summing Amplifier
- Lab: Integrator, Differentiator
- Feedback and Stability
- Lab: Oscillator
- Frequency Response
- Lab: Analog Filters
- Final Exam